Photovoltaic Data Preprocessing and Feature Engineering: An Empirical Strategy-to-Execution Audit

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***Abstract*—This reflection executes and audits a photovoltaic measurement preprocessing strategy using the released 1,373,798-row dataset. The workflow retains suspicious observations, isolates learned statistics to the training partition, and implements target-free transformations through Scikit-Learn Pipeline and ColumnTransformer. Eight diagnostic figures support numerical checks of missingness, distributions, collinearity, target association, scaling and exploratory PCA. The six-channel reference produces twelve mathematical inputs with availability indicators; alternative representations are documented without a predictive comparison. No classifier is fitted and no deployment-performance conclusion is drawn.**

***Index Terms*—Photovoltaic measurements, preprocessing, feature engineering, data quality, leakage prevention, mutual information, PCA.**

I. Scope, Data and Source Audit

The current milestone is data preprocessing and feature engineering. The original-author dataset [1] and associated monitoring study [2] provide two provisionally aligned source vectors. The immutable CSV preserves exported source order and contains $N = 1{,}373{,}798$ observations and $P = 7$ variables: six continuous predictors and nominal target f_nv (Table I). Voltage/current quantities are source-described DC RMS measurements; module temperature aggregates source probes. Integer target codes are labels, not an ordinal or continuous measurement.

The study describes nominal 1 Hz acquisition, per-second electrical RMS calculations, one-second irradiance/temperature averages and a 16-day experiment. Released arrays contain no timestamp. Absolute time and uninterrupted one-second spacing cannot be reconstructed. Sixteen signal-defined blocks are therefore not verified days, sessions or hours. All references to positions describe export order only.

The raw class counts are 1,162,931 / 5,999 / 10,371 / 6,024 / 188,473 for normal / short circuit / resistive degradation / open circuit / natural building shade. Filtering $irr \geq 100\,\mathrm{W/m^2}$ yields 515,958 rows and the literature counts 309,253 / 5,999 / 10,371 / 6,024 / 184,311 exactly. This suggests a possible evaluation-population explanation, but the study does not explicitly document that exact filter. Low-irradiance values are not all labeled nighttime, and no rows are removed to force agreement with the paper.

TABLE I
Raw schema and exported missingness (all rows).

Variable / role	Unit	Missing
`vdc1`, `vdc2`: continuous predictors	V	0 (0%)
`idc1`, `idc2`: continuous predictors	A	0 (0%)
`irr`: continuous predictor	$\mathrm{W/m^2}$	0 (0%)
`pvt`: continuous predictor	°C	0 (0%)
`f_nv`: nominal target	—	0 (0%)

II. Quality, Missingness and Outlier Decisions

A. Missingness and suspicious values

The implementation audits the approved Phase 1 strategy [3] against executed preprocessing evidence. Figure 1 displays a seed-42 uniform sample of 5,000 TRAIN positions. The full release has zero conventional missing cells out of 9,616,586 (0%); all 1,373,798 rows are complete. MCAR/MAR/MNAR inference from observed missingness is not applicable: neither Little's MCAR test nor missing-indicator Cramer's V is meaningful with a constant, all-observed mask. Unrecorded intervals remain unobservable. No missingness mechanism is inferred from zero NaNs.

The 62 paired $irr = 0, pvt = 0$ observations occur in runs of 1/32/29 and remain suspicious observations rather than confirmed missing values. The source does not establish zero as a missing sentinel. One irradiance value is $-0.0381\,\mathrm{W/m^2}$; the predeclared convention clamps $-1 \leq irr < 0$ to zero only in processed matrices. Values below -1 and negative electrical measurements would become unavailable. All 24,172 released negative-Celsius observations remain valid candidates; TRAIN temperature reaches $-1.6543\,°\mathrm{C}$ (Table II). The raw CSV remains unchanged.

TRAIN-median contingency imputation and per-retained-channel availability indicators are retained. There are no exported NaNs requiring ordinary imputation. A median is less sensitive to extremes than a mean, but a low-light median can still be unsuitable during generation. Mode is unsuitable for continuous measurements; KNN adds distance and computational assumptions; interpolation requires continuity not established here. Eligibility flags allow at most one unavailable retained channel for at most two consecutive source positions,

resetting at gaps/block boundaries; all arrays preserve row alignment.

B. Distribution-qualified screening rather than forced capping

Figure 3 compares raw with reference-cleaned TRAIN measurements. The audit uses $[Q_1 - 1.5\,IQR, Q_3 + 1.5\,IQR]$ and a complementary distribution-qualified $|Z| > 3$ screen. Table II reports the actual fences, skewness and unbiased Fisher excess kurtosis. Every predictor has zero IQR flags (0%) and zero $|Z| > 3$ flags under the retained full-TRAIN reference. Negative excess kurtosis and multi-regime mixtures do not justify a Gaussian assumption. In particular, current/irradiance mean–median differences show why no universal mean-filling rule is asserted.

No automatic statistical capping or row deletion is selected. Broad full-population fences are screens, not physical-validity certificates; rare fault-state measurements may carry genuine information. Nearly identical panels in Fig. 3 are expected, not evidence of a failed cleaning operation. The only observed quality change is the one near-zero irradiance clamp. This deterministic engineering convention changes one processed value and is not claimed to be literally lossless; its original value remains recoverable in the immutable export.

III. Relationships, Relevance and Feature Retention

Figure 2 shows all TRAIN labels at natural prevalence: 844,805 / 1,226 / 5,695 / 1,204 / 153,199, with normal representing 83.9659%. No target encoding as a continuous quantity, synthetic oversampling or row duplication is introduced. No categorical predictor requires ordinal or one-hot encoding. Nominal-target association is separated from predictor–predictor correlation.

Figure 4 is the upper-triangle six-predictor TRAIN Pearson matrix; f_nv is excluded. Table III lists all five pairs with $|r| \geq 0.85$. Figure 5 displays a uniform seed-42 sample of 20,000 TRAIN positions; correlations and descriptive OLS lines use full TRAIN. Spearman is a possible sensitivity method, not an unperformed result claimed in this report. Association does not imply significance or causation under dependent repeated readings.

The original TRAIN VIF values are independently checked as the diagonal of the inverse centered Pearson matrix, equivalent to intercept-OLS VIF. Five values exceed 10 (Table IV), confirming strong dependence without proving physical redundancy. All six reference inputs are retained; VIF/correlation alone does not select a physical-channel removal.

Figure 6 is *Target Association / Mutual Information Feature Screening*, not trained-model feature importance. Continuous-predictor/nominal-target MI uses a seed-42 proportional stratified sample of 50,000 TRAIN observations, $k = 3$, and TRAIN-fitted standardization. Sample counts are 41,983 / 61 / 283 / 60 / 7,613. Table IV gives the scores in nats. Imbalance, ties and dependent rows limit interpretation. These descriptive scores do not trigger predictor deletion or establish causal sensor importance.

IV. Scaling and Exploratory Dimensionality

Figure 7 compares the existing raw, StandardScaler, RobustScaler and MinMaxScaler irradiance representations. The standardized six-input reference supports scale-sensitive later methods; native-scale alternatives remain documented separately. Standardization aligns units/penalties rather than asserting Gaussian input. Table V records actual TRAIN behavior. Across all six standardized channels, absolute means are below 10^{-10} and population standard deviations are one within 10^{-10}.

The irradiance median is 1.8733 while the raw mean is 257.9285 W/m^2. Robust scaling centers the median at zero and maps its IQR to one; MinMax maps the quality-processed TRAIN range to [0,1]. Held-out values are not guaranteed to stay within [0,1]. Positive affine scaling leaves skewness effectively unchanged at 0.8745; the tiny raw-versus-processed difference reflects the single near-zero quality clamp. No log transform was selected or added merely to reproduce an illustrative assignment example. Robust/MinMax are diagnostic alternatives, not a retrospective replacement of the reference preprocessing.

Figure 8 includes individual/cumulative variance and both 90% and 95% cutoffs. Full-SVD PCA fits a TRAIN-only standardized six-column matrix; covariance eigenvalues are independently verified (Table VI). PC1 explains 82.5301%; the first two explain 95.1677%, crossing both cutoffs. A two-component diagnostic view omits 4.8323% of variance and is not described as lossless. PCA is not selected in the primary pipeline. All six measured inputs remain necessary to compute its components, so two mathematical components do not mean two physical sensors.

V. Leakage-Free Architecture and Measured Outputs

The outer Pipeline is `RetainedQuality` → `ColumnTransformer` [4] (Table VIII). Deterministic quality checks sit before the ColumnTransformer because physical conventions apply to retained-channel DataFrames. Its numeric branch contains median imputation, optional parent-constrained physical features and optional StandardScaler; the parallel branch supplies all retained-channel missingness indicators. No target or source/block identifier enters either branch. Diagnostic Robust/MinMax and StandardScaler–PCA chains are also explicit Pipelines; their learned parameters use TRAIN only.

The ordered partitions retain 1,006,129 / 116,977 / 146,461 rows; separator intervals [1006129,1056990) and [1173967,1227337) withhold 104,231 source positions. TRAIN is [0,1006129), validation [1056990,1173967), and test [1227337,1373798). These rows remain in the raw export. The split follows 12/2/2 signal-defined blocks; the separator rule $irr < 100\,\mathrm{W/m^2}$ for at least 3,600 consecutive positions does not establish a verified hour. Neighbor dependence precludes a naive random row split. The original validation sub-block purposes remain planning metadata, not operations executed in this preprocessing submission.

TABLE II
TRAIN DISTRIBUTION AND OUTLIER AUDIT, BEFORE REFERENCE CLEANING. $N = 1{,}006{,}129$ PER PREDICTOR. ALL IQR AND $|Z| > 3$ COUNTS ARE ZERO; EXCESS KURTOSIS IS UNBIASED FISHER KURTOSIS.

Predictor	Q_1	Q_3	IQR	Lower	Upper	Skew	Excess kurt.
vdc1	1.0184	274.4720	273.4536	-409.1620	684.6524	0.0683	-1.9302
vdc2	0.9824	272.3780	271.3956	-406.1110	679.4714	0.0909	-1.8973
idc1	0.0648	4.9670	4.9022	-7.2885	12.3203	1.0046	-0.6707
idc2	0.0088	4.0011	3.9923	-5.9797	9.9896	1.1450	-0.3666
irr	1.2889	586.3810	585.0921	-876.3492	1464.0192	0.8745	-0.9456
pvt	12.5516	34.0299	21.4783	-19.6658	66.2473	0.5157	-0.6303

TABLE III
FIVE FLAGGED TRAIN PREDICTOR PAIRS. FULL-TRAIN VALUES SUPPORT FIGS. 4–5.

Predictors	Pearson r
vdc1 / vdc2	0.989103
idc1 / idc2	0.952833
idc1 / irr	0.979472
idc1 / pvt	0.862289
idc2 / irr	0.955982

TABLE IV
TRAIN-ONLY VIF VERIFICATION AND DESCRIPTIVE NOMINAL-TARGET MI. MI USES A PROPORTIONAL 50,000-OBSERVATION SAMPLE, NOT MODEL-BASED IMPORTANCE.

Predictor	VIF	MI (nats)
vdc1	52.168202	0.178615
vdc2	56.345662	0.220611
idc1	31.495610	0.276227
idc2	15.780847	0.276981
irr	39.292651	0.300592
pvt	4.069798	0.112375

The guarded fit entry point checks the complete ordered TRAIN index before constructing a preprocessor. A held-out fit attempt is rejected before the factory is called. Medians/scaling/PCA parameters and the fit ledger are checked against TRAIN reference evidence; validation/test are transformed only. The historical reserved partition is not claimed to be previously unseen or pristine; no downstream evaluation is presented.

A. Initial and final $N \times P$

The full raw matrix is $1{,}373{,}798 \times 7$, including the separate nominal target. The predictor matrix initially has six columns. Table VII measures each stage for all three partitions. The primary six-channel reference retains every predictor and partition row; six availability flags produce twelve mathematical design columns. No missing exported measurement is synthesized. Scaling preserves numerical column count, and no information-loss claim is made for the single documented irradiance convention.

The electrical-native and electrical-derived branches are existing alternative representations, not evidence that four channels suffice. Four raw values plus four flags yield eight columns. The derived branch adds $P_1 = vdc1\,idc1$, $P_2 = vdc2\,idc2$, $P_{total} = P_1 + P_2$, $|vdc1 - vdc2|$ and $|idc1 - idc2|$, giving thirteen mathematical columns. RMS products are proxies, not independent instantaneous-power measurements. Every derived feature uses retained parents. PCA remains a separate diagnostic $N \times 6$ to $N \times 6$ transformation, with an optional $N \times 2$ inspection view; it does not replace the primary $N \times 12$ design.

VI. STRATEGY VERSUS REALITY AND REPRODUCIBILITY

Tables IX–X compare meaningful Phase 1 preprocessing choices [3] with actual execution and empirical evidence. Retained decisions are identified directly; no deviation is invented. MI/PCA and alternative representations are descriptive implementation additions, not downstream predictive conclusions.

The executed notebook contains thirteen sequential code cells. It reads the immutable export, checks schema/quality, fits only preprocessing/diagnostic estimators on TRAIN, transforms held-out arrays, and records compact measured evidence. All eight existing 300-DPI PNGs are retained byte-for-byte in `figures/`; no cosmetic regeneration occurred. Reusable transformers, pipeline construction and statistical audit code reside in `src/`. The report source uses the unmodified IEEEtran conference class [5]; repository instructions distinguish notebook execution, evidence verification and PDF compilation. No private review files, environment, model cache or prediction arrays are required.

VII. LIMITATIONS AND CURRENT-STAGE CONCLUSION

One controlled Brazilian plant does not establish geographic or deployment transferability. Neighboring rows are not independent fault episodes; uncertain timing and provisional paired-vector alignment remain limitations. Zero NaNs cannot identify omitted intervals. Full-TRAIN statistical screens may not detect condition-specific anomalies, so zero IQR flags are not a safety guarantee. MI/PCA describe association/variance rather than causality or sensor adequacy. Duplicate removal is not claimed. The output is an empirically justified, reproducible preprocessing implementation; no classifier recommendation, model performance or resource-acceptance result is reported.

TABLE V
TRAIN IRRADIANCE SCALING BEHAVIOR (POPULATION STANDARD DEVIATION). VALUES FOLLOW QUALITY PROCESSING EXCEPT THE RAW ROW.

Representation	Mean	Std.	Median	Min.	Max.	Skew
raw	257.928481	349.018335	1.873300	-0.038100	1085.920000	0.874462
standard	-0.000000	1.000000	-0.733644	-0.739011	2.372344	0.874462
robust	0.437632	0.596519	0.000000	-0.003202	1.852780	0.874462
minmax	0.237521	0.321403	0.001725	0.000000	1.000000	0.874462

TABLE VI
EXPLORATORY TRAIN PCA: SIX STANDARDIZED PHYSICAL INPUTS. RATIOS ARE FRACTIONS.

PC	Eigenvalue	Individual	Cumulative
1	4.951809	0.825301	0.825301
2	0.758256	0.126376	0.951677
3	0.214540	0.035757	0.987433
4	0.049972	0.008329	0.995762
5	0.016539	0.002756	0.998518
6	0.008889	0.001482	1.000000

TABLE VII

MEASURED PREDICTOR STAGES FOR EVERY PARTITION. N STAYS FIXED ACROSS EACH ROW. RAW $P = 7$ INCLUDES THE EXCLUDED TARGET; ORIGINAL PREDICTOR $P = 6$. M+I LISTS NUMERIC MEDIANS PLUS THE PARALLEL INDICATORS; C IS THEIR LOGICAL CONCATENATION BEFORE OPTIONAL DERIVED FEATURES. F AND S COUNT NUMERIC COLUMNS AFTER FEATURE ENGINEERING AND SCALING; D IS THE ACTUAL FINAL CONCATENATION. NATIVE/DERIVED POLICIES APPLY NO NUMERICAL SCALER. PCA DIAGNOSTIC: SIX INPUT/ALL COMPONENTS, TWO-COMPONENT 95% VIEW; PRIMARY DESIGN UNCHANGED.

Representation	Partition	N	P_0	Phys.	Q	M+I	C	F	S	D
Six standardized (reference)	TRAIN	1,006,129	6	6	6	6+6	12	6	6	12
Six standardized (reference)	VALIDATION	116,977	6	6	6	6+6	12	6	6	12
Six standardized (reference)	TEST	146,461	6	6	6	6+6	12	6	6	12
Electrical native (alternative)	TRAIN	1,006,129	6	4	4	4+4	8	4	4	8
Electrical native (alternative)	VALIDATION	116,977	6	4	4	4+4	8	4	4	8
Electrical native (alternative)	TEST	146,461	6	4	4	4+4	8	4	4	8
Electrical derived (alternative)	TRAIN	1,006,129	6	4	4	4+4	8	9	9	13
Electrical derived (alternative)	VALIDATION	116,977	6	4	4	4+4	8	9	9	13
Electrical derived (alternative)	TEST	146,461	6	4	4	4+4	8	9	9	13

TABLE VIII

ACTUAL LEAKAGE-FREE PREPROCESSING ARCHITECTURE. TRAIN $N = 1{,}006{,}129$ FOR EVERY LEARNED FIT; HELD-OUT PARTITIONS TRANSFORM ONLY.

Location / component	Operation and fitting scope
Outer Pipeline: quality	Deterministic schema/finite/physical conventions; no estimated distribution parameters.
ColumnTransformer: numeric	TRAIN median; optional target-free derivatives; optional TRAIN StandardScaler.
ColumnTransformer: flags	TRAIN MissingIndicator(features=all); one indicator per retained physical input.
Parallel diagnostics	Robust/MinMax Pipeline and StandardScaler–PCA Pipeline; full TRAIN fits only.
Guard / assembly	TRAIN-only schema guard, separate nominal targets, row-aligned held-out transforms.

TABLE IX

PREPROCESSING STRATEGY VERSUS REALITY. RETENTION IS NOT DESCRIBED AS A FORCED DEVIATION. FIGURE NUMBERS REFER TO THIS IEEE PAPER.

Step	Phase 1 planned strategy	Phase 2 actual execution	Empirical justification / metric / figure
Missingness / imputation	TRAIN-median contingency and availability indicators	Retained; no exported gaps to fill	0/9,616,586 missing cells (0%); Fig. 1; Table I
Suspicious paired zeros	Annotate, without assuming a missing sentinel	Retained as observations, not relabeled missing	62 paired irr/pvt zeros; runs 1/32/29; Fig. 1
Negative irradiance	Near-zero physical convention with caution	Retained: -1 <= irr < 0 becomes zero in processed data	One value -0.0381 W/m^2; no raw CSV change; Fig. 3
Negative temperature	Retain physically valid negative Celsius	Retained; no automatic zero-floor	24,172 released negative-Celsius rows; TRAIN min -1.6543 °C; Fig. 3
Outlier policy	IQR / distribution-qualified Z screening; no automatic capping	Retained; no winsorization or row deletion	All six TRAIN IQR counts=0 (0%); skew 0.0683–1.1450; Table II; Fig. 3
Target / imbalance	Nominal f_nv; no categorical predictor encoding or initial SMOTE	Retained observed labels and natural partition prevalence	TRAIN counts 844805/1226/5695/1204/153199; normal 83.966%; Fig. 2
Predictor correlation	TRAIN Pearson |r| >= 0.85; assess physical meaning before deletion	Retained screen; no automatic predictor removal	Five flagged pairs, r=0.8623–0.9891; Figs. 4–5; Table III
VIF / retention	TRAIN VIF >10 warning; preserve reference physical inputs	Retained; all six reference channels remain	Five VIF>10; maximum 56.3457; Table IV; Figs. 4–5
Target relevance	TRAIN-only MI for continuous predictors / nominal target	Implemented descriptive proportional-sample MI; no feature deletion	N=50,000, k=3, seed=42; MI 0.1124–0.3006 nats; Fig. 6; Table IV

TABLE X

PREPROCESSING STRATEGY VERSUS REALITY (CONTINUED). RETENTION IS NOT DESCRIBED AS A FORCED DEVIATION. FIGURE NUMBERS REFER TO THIS IEEE PAPER.

Step	Phase 1 planned strategy	Phase 2 actual execution	Empirical justification / metric / figure
Standardization	TRAIN-fitted standardization for scale-sensitive representations	Retained six-channel standardized reference; flags unscaled	Max absolute TRAIN scaled mean 2.02e-16; each population std=1; Fig. 7; Table V
Scaling alternatives	Robust/MinMax sensitivity; optional justified log transform	Robust/MinMax diagnostic only; no log transform selected	irr robust median=0, IQR=1; MinMax TRAIN range=[0,1]; skew stays 0.8745; Fig. 7
Exploratory PCA	TRAIN-standardized variance/loadings; not physical sensor reduction	Implemented six-component diagnostic; reference pipeline not reduced	PC1=82.5301%; first two=95.1677%; 90%/95% need two; Fig. 8; Table VI
Ordered split	12/2/2 signal-block split and withheld separators	Retained exact bounds; no random row split or fabricated time	N=1006129/116977/146461; 104231 separators; Table VII
Leakage controls	Fit learned quantities on TRAIN; transform held-out only	Retained; guarded entry point rejects held-out fit before factory call	All fit-ledger N=1006129; TRAIN digest verified; Table VIII
Physical features	Domain-grounded features with retained-parent constraints	Existing optional electrical-derived representation retained; not a sensor-selection outcome	4 raw + 5 derived + 4 flags=13 columns; all parents retained; Table VII
Output dimensions	Track retained observations and physical/mathematical inputs	Measured every quality, median, flag, feature and scaling stage	Primary: N×6 -> N×12; alternatives N×8 / N×13; Table VII
Population discrepancy	Verify source/count mismatch without forcing agreement	Retained full raw population; descriptive irradiance-qualified audit	irr >=100 gives N=515958 and literature counts exactly; source does not specify exact filter; Table I

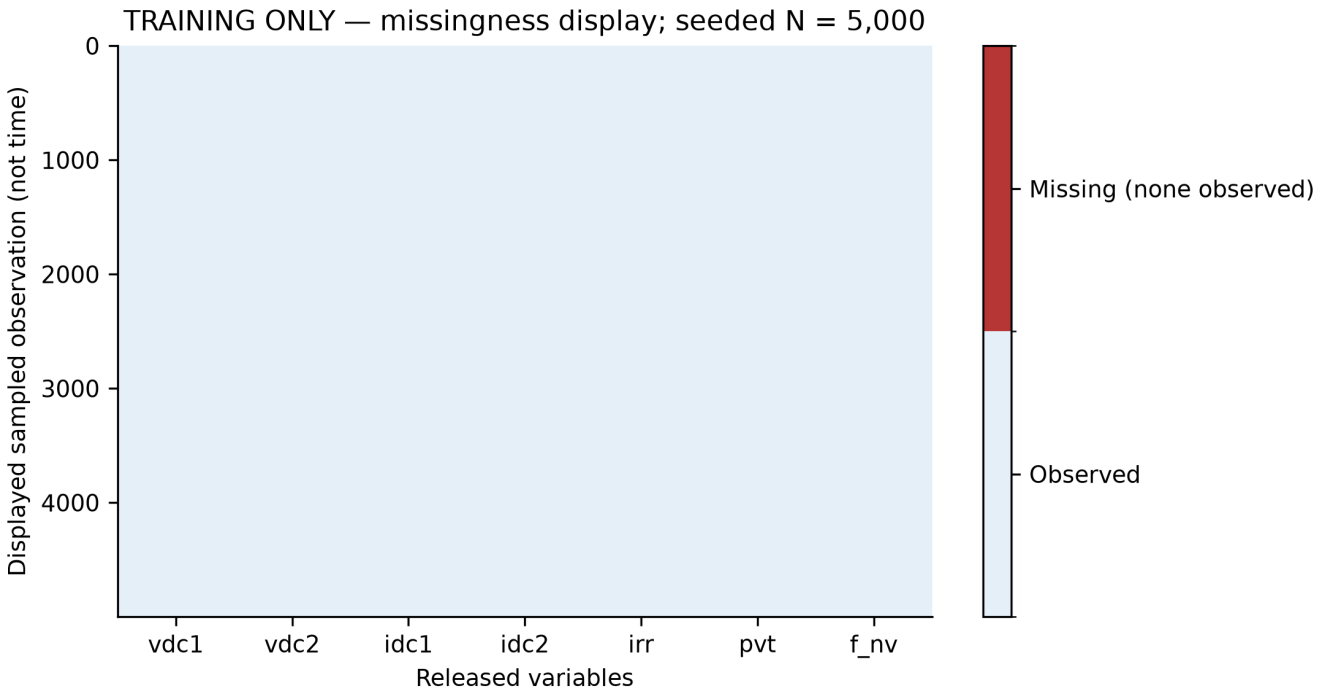


Fig. 1. Missingness pattern: seed-42 uniform TRAIN display, N=5,000. Full-export conventional missingness is 0/9,616,586 cells; the 62 paired zeros remain observations.

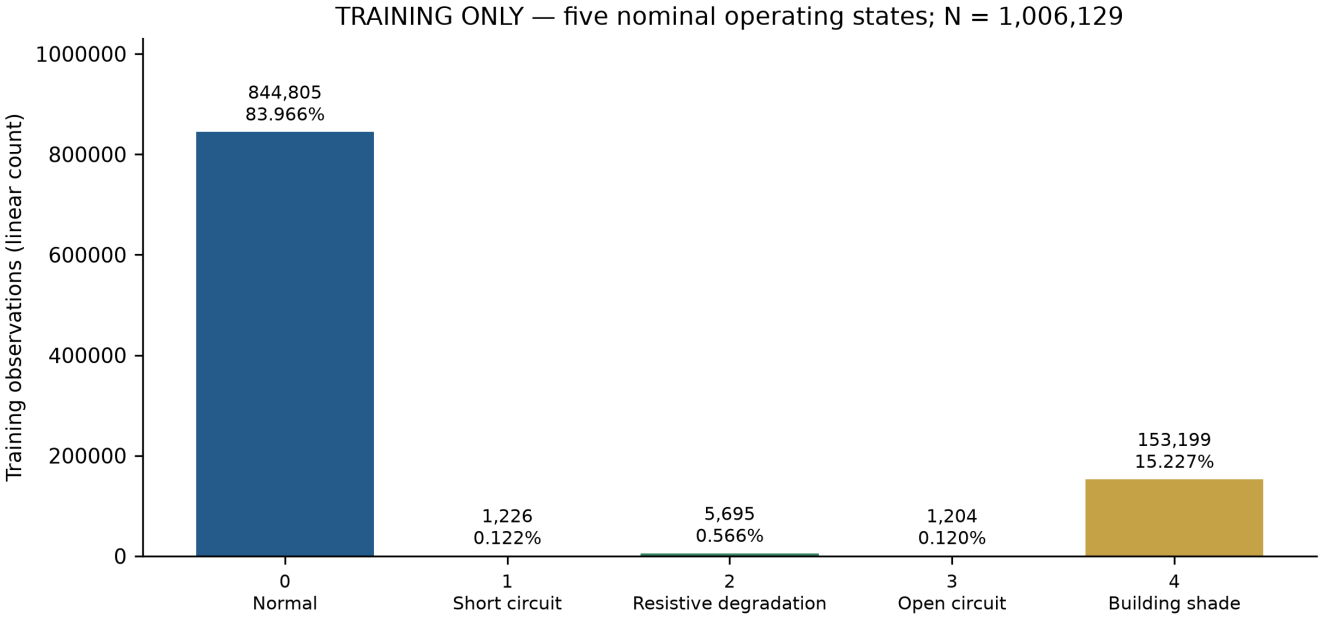


Fig. 2. Target distribution: all 1,006,129 TRAIN positions, unchanged nominal class labels and natural prevalence. Normal contributes approximately 83.966%; no resampling is applied.

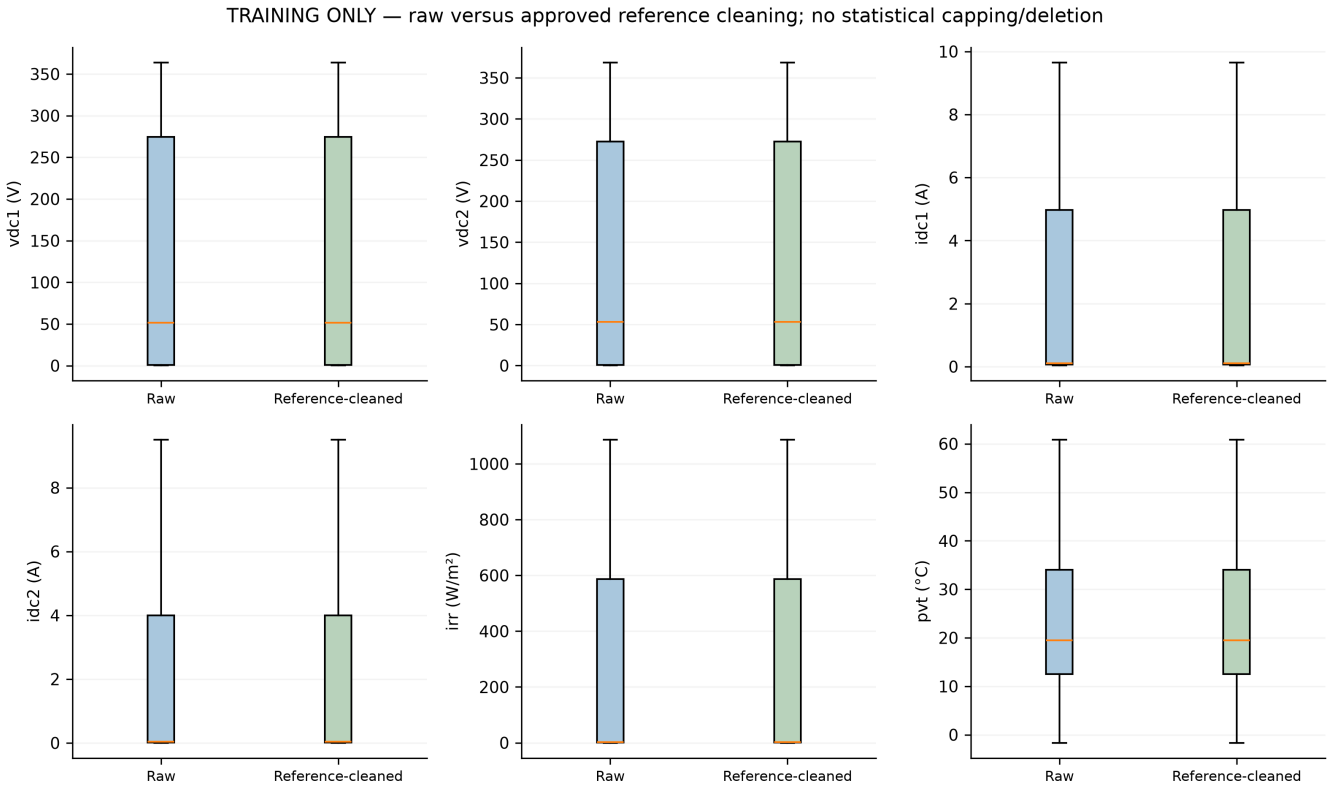


Fig. 3. TRAIN raw versus reference-cleaned boxplots with IQR whiskers and retained observations. No statistical capping or deletion is selected; zero outside-fence flags reflect the actual full-TRAIN screen, not physical certification.

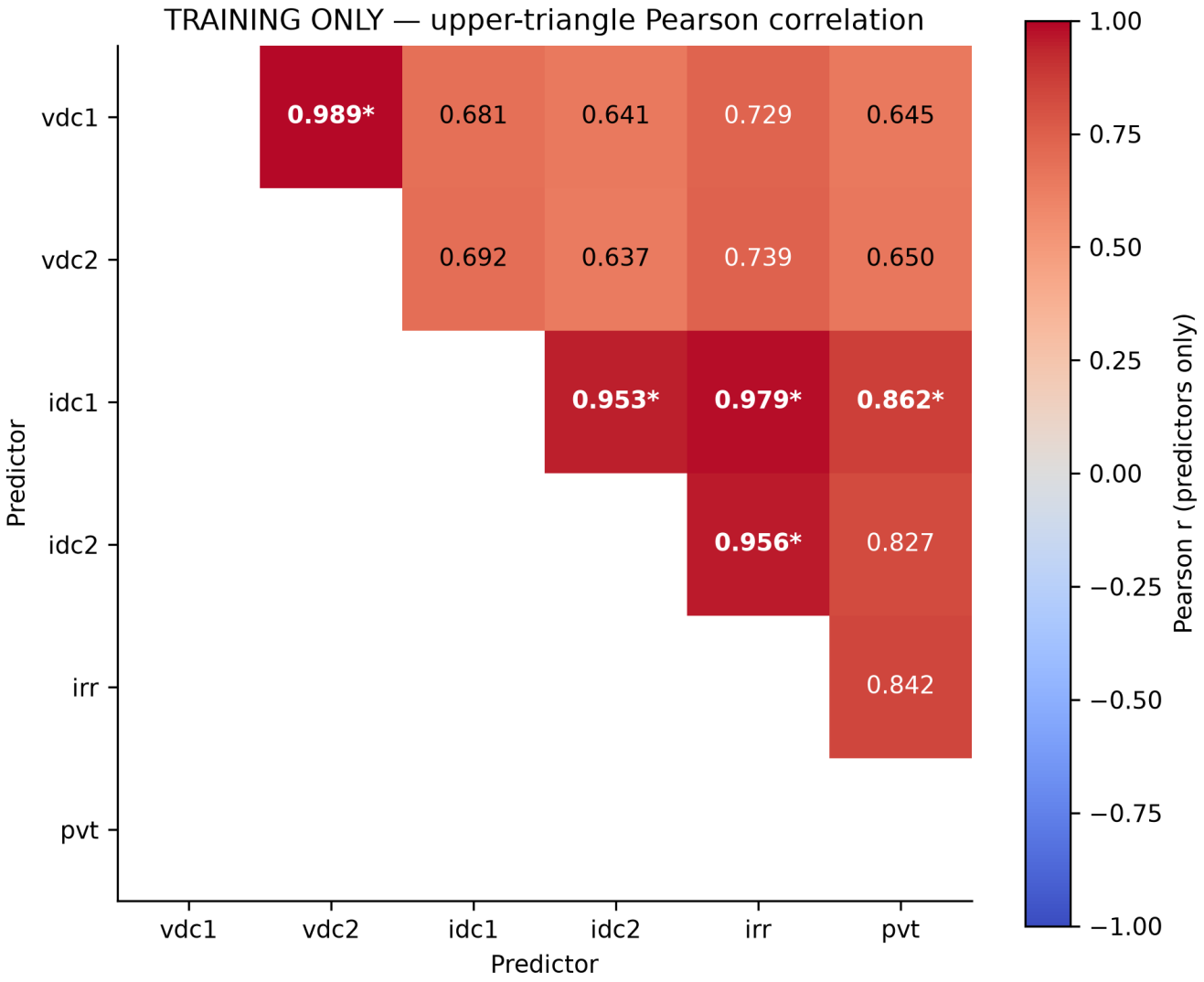


Fig. 4. TRAIN predictor-only upper-triangle Pearson correlation. Five pairs satisfy $|r| \geq 0.85$; f_nv is excluded. Warning thresholds do not automatically remove physical measurements.

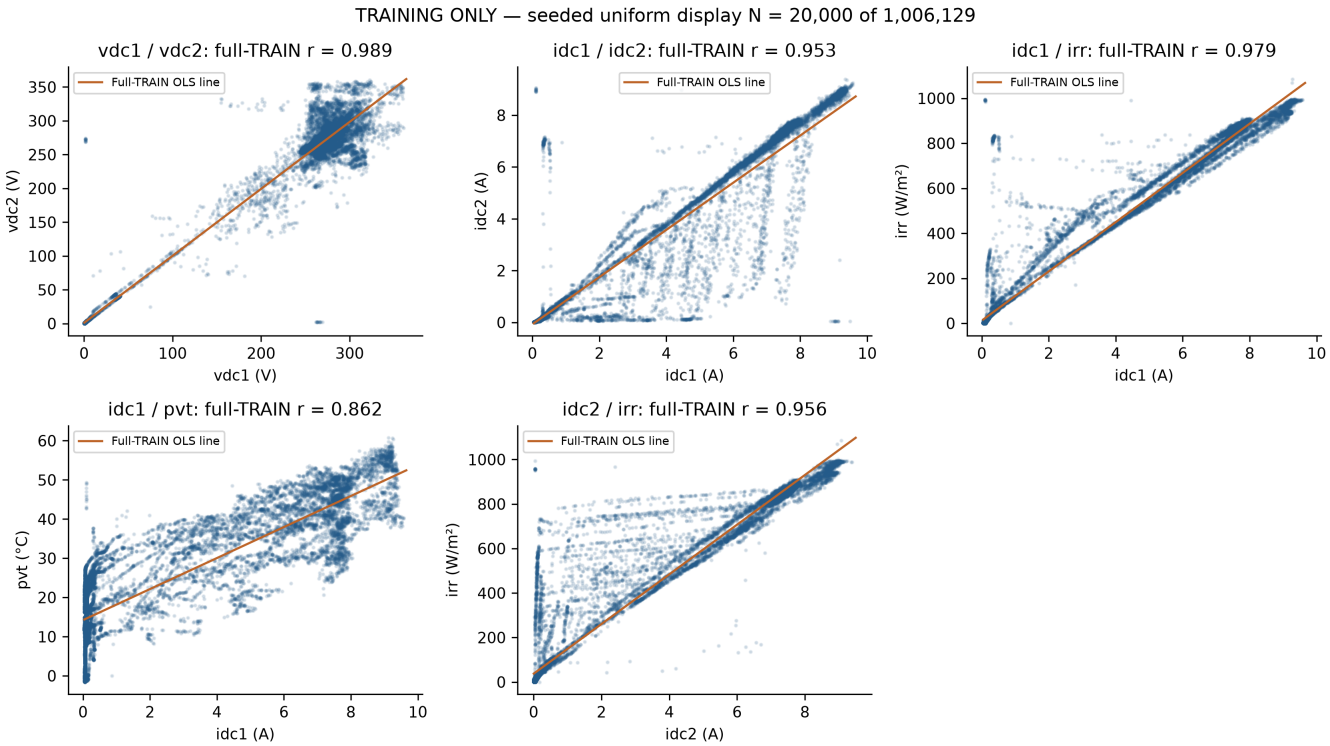


Fig. 5. All five collinear pairs. Seed-42 uniform TRAIN scatter display N=20,000; fitted descriptive lines and Pearson values use full TRAIN. Neither lines nor association imply causal or independent samples.

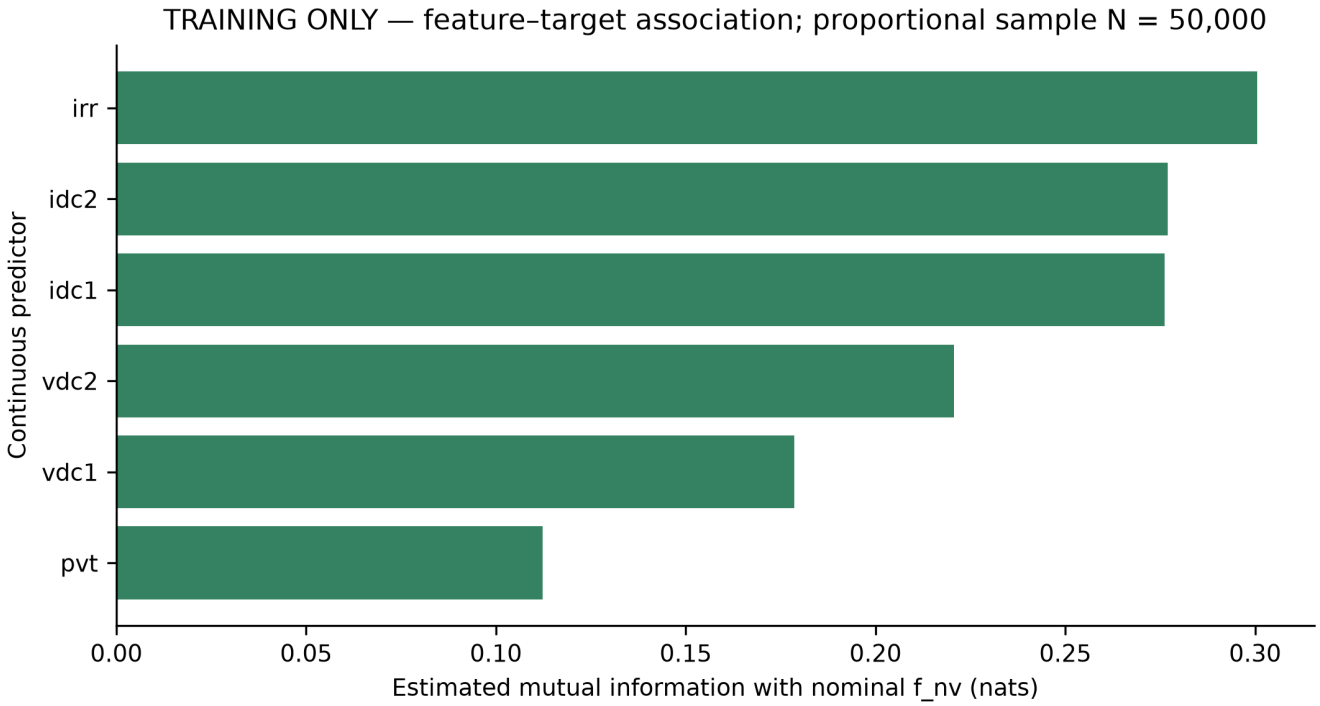


Fig. 6. Target Association / Mutual Information Feature Screening. Continuous-predictor / nominal-target MI, seed 42, proportional TRAIN sample N=50,000, k=3. This is descriptive preprocessing relevance, not trained-model feature importance.

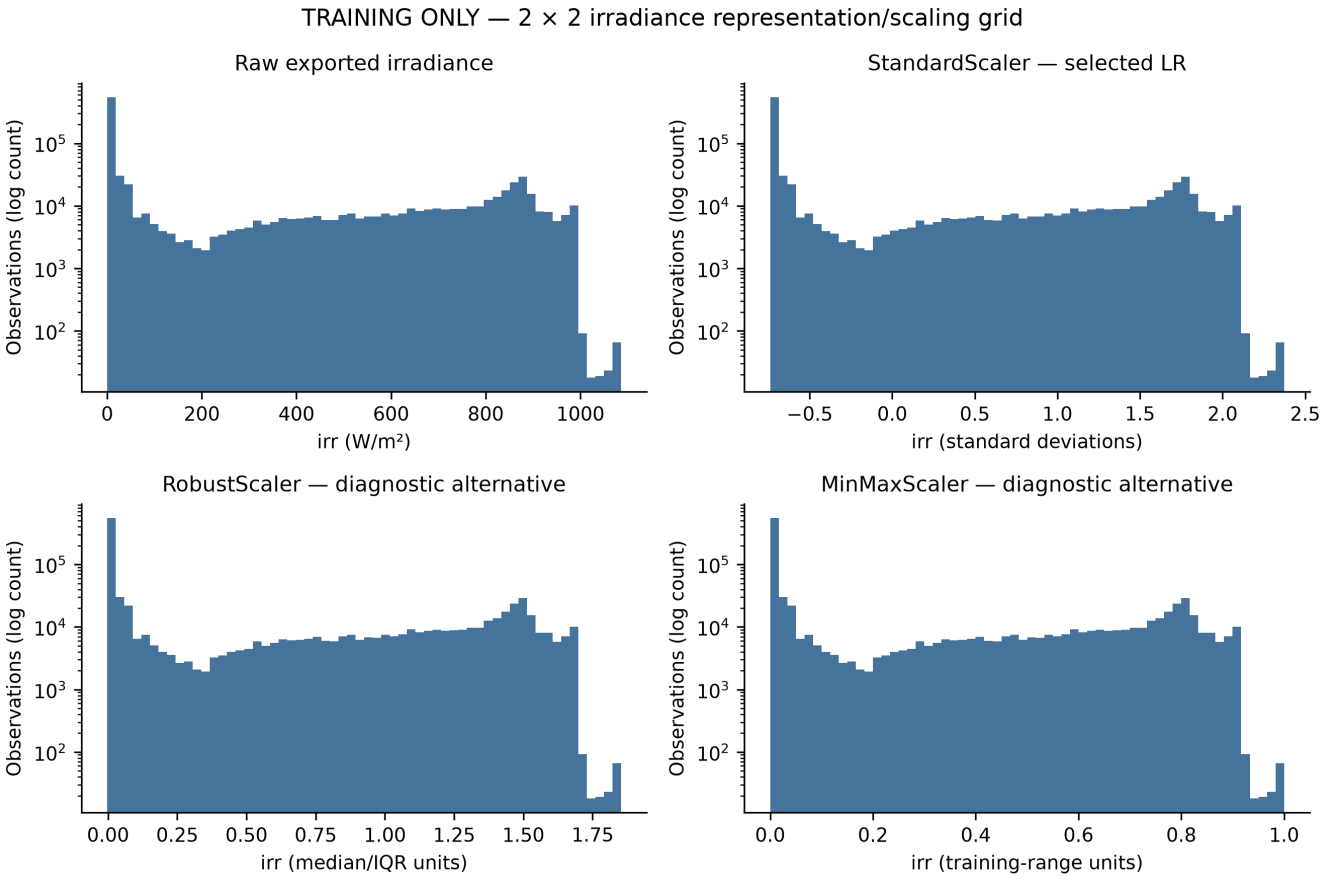


Fig. 7. Original TRAIN transformation/scaling grid. StandardScaler is the reference standardized representation; Robust/MinMax are diagnostics. Original algorithm labels describe intended preprocessing rationale only. No log transform or downstream model-selection outcome is claimed.

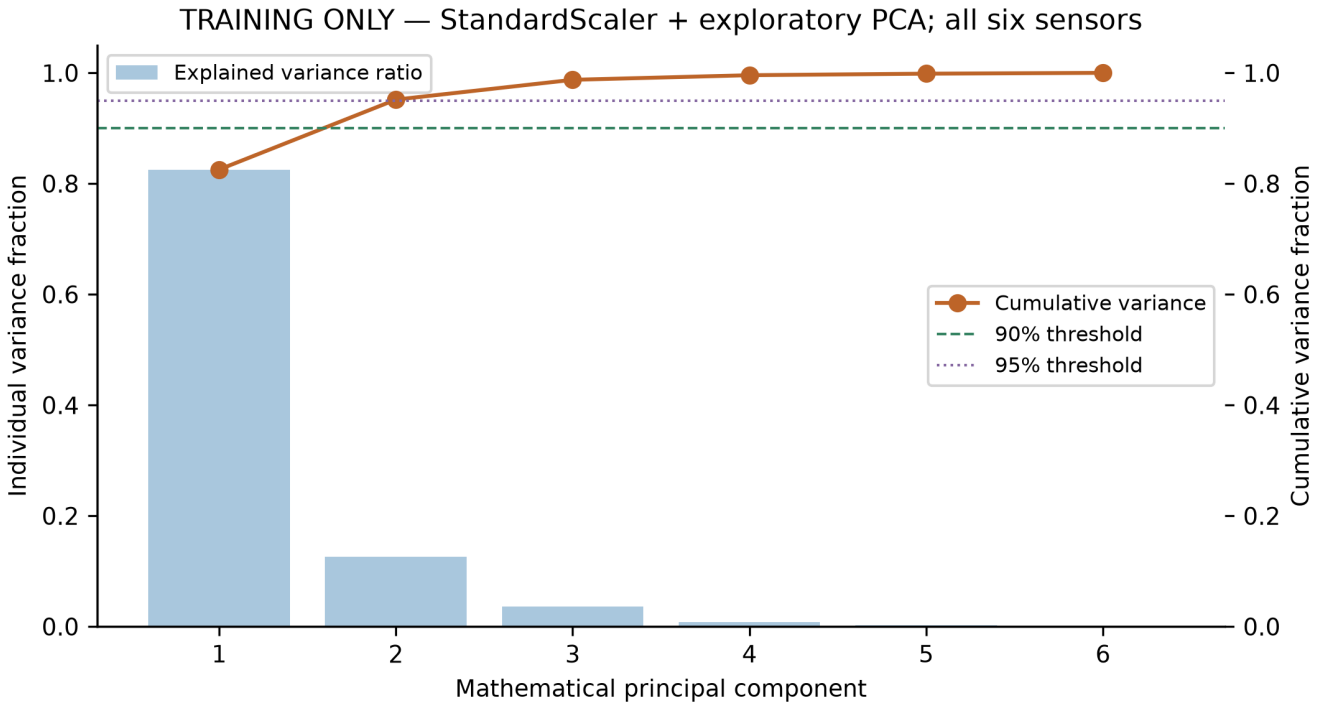


Fig. 8. Exploratory StandardScaler–PCA fitted on full TRAIN only. Bars give individual variance, line cumulative variance, dashed/dotted lines 90%/95% cutoffs. Two components explain 95.1677%, requiring all six physical inputs.

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